Ollie CISA Coach

Step-by-step guide to hosting your own copy online, with Google sign-in, synced progress and a leaderboard

For the package **ollie-host.zip** · October 2026

What you are setting up

```
Browser ──► Caddy (HTTPS) ──► Node app ──► SQLite file (data/coach.db)
                                  └─ checks Google sign-in tokens
```

Visitors open your website, sign in with Google, and their study progress is saved on your server so it follows them across devices. People who choose to can join a leaderboard that shows only a nickname they type, never their name or email.

> **Time needed:** about 1 to 2 hours the first time. **Cost:** about $10 per year for a domain plus $4 to $6 per month for a small server.

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1. Before you start

You need	Notes
A domain name	For example from Namecheap, Cloudflare or GoDaddy. This guide uses `coach.example.com`; replace it with yours everywhere.
A Google account	To create the sign-in credentials. Turn on 2-step verification.
A small Linux server (VPS)	Hetzner, DigitalOcean, AWS Lightsail or Vultr. 1 CPU and 1 GB of memory is enough. Choose **Ubuntu 24.04**.
The package	`ollie-host.zip`, unzipped on your computer.
A terminal	Terminal on Mac, or PowerShell on Windows, to connect with `ssh`.

2. Create Google sign-in credentials (about 10 minutes)

2.1 Create a project

1. Go to **console.cloud.google.com** and sign in.
2. Click the project picker at the top, then **New project**. Name it `Ollie CISA Coach` and click **Create**. Make sure it is selected.

2.2 Set up the consent screen

1. Open **APIs & Services → OAuth consent screen** (in the newer console: **Google Auth Platform → Branding**).
2. User type: **External**, then **Create** or **Get started**.
3. App name: `Ollie CISA Coach`. Add your email as support and developer contact.
4. App domain: home page `https://coach.example.com`, privacy policy `https://coach.example.com/privacy.html`, terms `https://coach.example.com/terms.html`. Add `example.com` under Authorized domains.
5. Scopes: keep only the defaults (openid, email, profile). These do not need Google's review.
6. Click **Publish app** so it is **In production**. In "Testing" mode only listed test users can sign in.

2.3 Create the Client ID

1. **APIs & Services → Credentials → Create credentials → OAuth client ID**.
2. Application type: **Web application**, name `Ollie web`.
3. Under **Authorized JavaScript origins** add `https://coach.example.com` exactly (no slash at the end). Add the `www` version too if you use it.
4. Leave **Authorized redirect URIs** empty, then click **Create**.
5. Copy the **Client ID**; it ends in `.apps.googleusercontent.com`. There is no client secret to keep.

> Changes to origins can take from a few minutes to a few hours to work.

3. Rent a server and point your domain at it

Good choices for India

Pick a data centre in India (Mumbai, Bengaluru, Delhi or Hyderabad) so the site is fast for Indian users. Any of these works with this guide; choose Ubuntu 24.04 and the smallest plan with 1 GB of memory.

Provider	India locations	Why pick it	Rough cost
DigitalOcean (recommended)	Bengaluru	Simplest dashboard, good guides, built-in firewall and backups	about $6/month
AWS Lightsail	Mumbai	Fixed monthly price, Amazon reliability, easy snapshots	about $5/month
Vultr	Mumbai, Delhi, Bengaluru	Most Indian locations, hourly billing	about $5 to $6/month
Akamai (Linode)	Mumbai, Chennai	Simple and reliable	about $5/month
Hostinger VPS	Mumbai	Pay in rupees, low prices on long plans	a few hundred ₹/month
E2E Networks	Delhi NCR, Mumbai	Indian company, GST invoices, data stays in India	a few hundred ₹/month
Oracle Cloud Free Tier	Mumbai, Hyderabad	Always-free server, good for trying it out; sign-up and capacity can be tricky	free

Prices change often and may add 18% GST; check the provider's site. Hetzner, often the cheapest elsewhere, has no India location (its nearest is Singapore). For the domain, a `.in` or `.com` from Cloudflare, Namecheap, GoDaddy or Hostinger costs roughly ₹700 to ₹1,200 a year.

1. Create an **Ubuntu 24.04** server at your provider and note its **IP address**. Add your SSH key, or note the root password.
2. At your domain registrar, open DNS settings and add an **A record**: name `coach`, value = the server's IP address. Use `@` as the name for the bare domain.
3. Wait until `ping coach.example.com` on your computer shows the server's IP. This usually takes 5 to 30 minutes.
4. If your provider has a firewall panel, allow ports **22**, **80** and **443**.

4. Install and start the app (Docker, recommended)

4.1 Connect and prepare the server

```
ssh root@YOUR_SERVER_IP
curl -fsSL https://get.docker.com | sh
ufw allow OpenSSH && ufw allow 80 && ufw allow 443 && ufw enable
```

4.2 Upload the package

From your own computer, in the folder that holds the zip:

```
scp ollie-host.zip root@YOUR_SERVER_IP:~/
```

Then on the server:

```
apt-get install -y unzip
unzip ollie-host.zip && cd cisa-coach-host
```

4.3 Fill in your settings

```
cp .env.example .env
openssl rand -hex 32        # copy the result: this is your session secret
nano .env
```

Fill in at least these, then save with Ctrl+O, Enter, Ctrl+X:

Setting	What to put
`BASE_URL`	`https://coach.example.com` exactly as people will type it, no slash at the end
`GOOGLE_CLIENT_ID`	The Client ID from step 2.3
`SESSION_SECRET`	The long random value from `openssl rand -hex 32`
`ADMIN_EMAILS`	Your Google email, for the admin page
`OPERATOR_NAME`, `CONTACT_EMAIL`	Your name and a contact email, shown in the privacy notice and terms

4.4 Set your domain for HTTPS

```
nano Caddyfile      # replace coach.example.com with your domain
```

Caddy gets and renews the HTTPS certificate for you automatically.

4.5 Start it

```
mkdir -p data && chown -R 1000:1000 data
docker compose up -d --build
docker compose logs -f app       # wait for "listening on :3000", then Ctrl+C
```

> If the build fails with **429 Too Many Requests** from Docker Hub, use a mirror:
>
> ```
> docker compose build --build-arg NODE_IMAGE=mirror.gcr.io/library/node:22-bookworm-slim
> docker compose up -d
> ```

5. Check it works

1. Open `https://coach.example.com`. The first load can take a few seconds while the certificate is issued.
2. Click **Sign in** at the top right, then **Continue with Google**.
3. Answer a quiz. Open the site on another device and sign in: your progress should be there.
4. Go to **More → Study buddies**, pick a nickname and **Join the board**. A second Google account should see it.
5. Open `https://coach.example.com/admin.html` while signed in with your admin email to see counts and moderate the board.
6. Health check: `curl https://coach.example.com/healthz` should return `{"ok":true}`.

6. Backups and updates

Daily backups (set up on day one)

```
docker compose exec app node scripts/backup.js     # writes data/backups/, keeps 14
```

Run it daily with `crontab -e` and this line:

```
15 3 * * * cd /root/cisa-coach-host && docker compose exec -T app node scripts/backup.js
```

> A backup on the same server is not a real backup. Copy `data/backups/` elsewhere regularly, for example to your laptop with `scp` or to cloud storage with `rclone`.

Restore: stop the app (`docker compose stop app`), copy a backup over `data/coach.db`, delete `data/coach.db-wal` and `data/coach.db-shm`, then `docker compose start app`.

Install a new version of the coach

```
# upload the new cisa-coach.html into app-source/, then on the server:
docker run --rm -v "$PWD":/w -w /w mirror.gcr.io/library/node:22-bookworm-slim \
  node scripts/import-app.js app-source/cisa-coach.html
docker compose up -d --build
```

The page is built into the app image, so the rebuild step is needed. People get the new version the next time they load the page, and their progress is kept.

Update the server software

```
docker compose up -d --build
```

7. Other ways to host

Option	When to use it
VPS without Docker	Install Node 22 and Caddy, then use `deploy/ollie.service` (systemd) and `deploy/Caddyfile.nodocker`. The full commands are in README section 2, Option B.
Render, Railway or Fly.io	Works with the Dockerfile, but you must add a **persistent disk** mounted at `/app/data`, usually a paid add-on. Without it, accounts are wiped on every deploy. Set the `.env` values in their dashboard and set `TRUST_PROXY=1`.
GitHub Pages, Netlify (static only)	Not suitable for sign-in or the leaderboard. You could host only `public/index.html` as a single-device app.

8. Troubleshooting

What you see	Cause and fix
No Sign in button	`GOOGLE_CLIENT_ID` is not set in `.env`. Fix it and run `docker compose up -d`.
Google says "origin_mismatch"	Your exact address is not in Authorized JavaScript origins. Check https, no trailing slash, and www vs no www. Wait a few minutes after changing it.
"Sign-in failed" after choosing an account	The Client ID in `.env` differs from Google's, or the server clock is wrong.
Signed in, but sync or leaderboard fails	`BASE_URL` does not exactly match the address in the browser.
Everyone is signed out after a restart	`SESSION_SECRET` is missing or was changed.
"Cannot open database" or EACCES	Run `chown -R 1000:1000 data`.
No HTTPS certificate	DNS doesn't point to the server yet, or ports 80 and 443 are blocked. Check with `docker compose logs caddy`.

9. Before you share the link publicly

- **Privacy notice and terms:** templates are at `/privacy.html` and `/terms.html`. Have them reviewed for your country (for example India's DPDP Act or GDPR). They are not legal advice.
- **Content:** keep the ISACA non-affiliation footer, and do not add text copied from ISACA's Review Manual or QAE database.
- **Off-site backups** and free **uptime monitoring** (UptimeRobot or Better Stack on `/healthz`).
- **Security:** 2-step verification on your Google and server-provider accounts. Never set `DEV_FAKE_AUTH` on a real server.

- **India:** if you collect personal data from Indian users, the DPDP Act 2023 applies; picking an India data centre keeps the data in the country.
- Optional: analytics stay off unless you set `ANALYTICS_SRC`, and then visitors are asked first.

> **Only you can do these:** buy the domain, create the server, create the Google credentials, fill in `.env`, and test a real Google sign-in on your domain. The package itself was tested before delivery: Docker build, HTTPS through Caddy, Google token checks, sync, leaderboard, data download and account deletion.

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